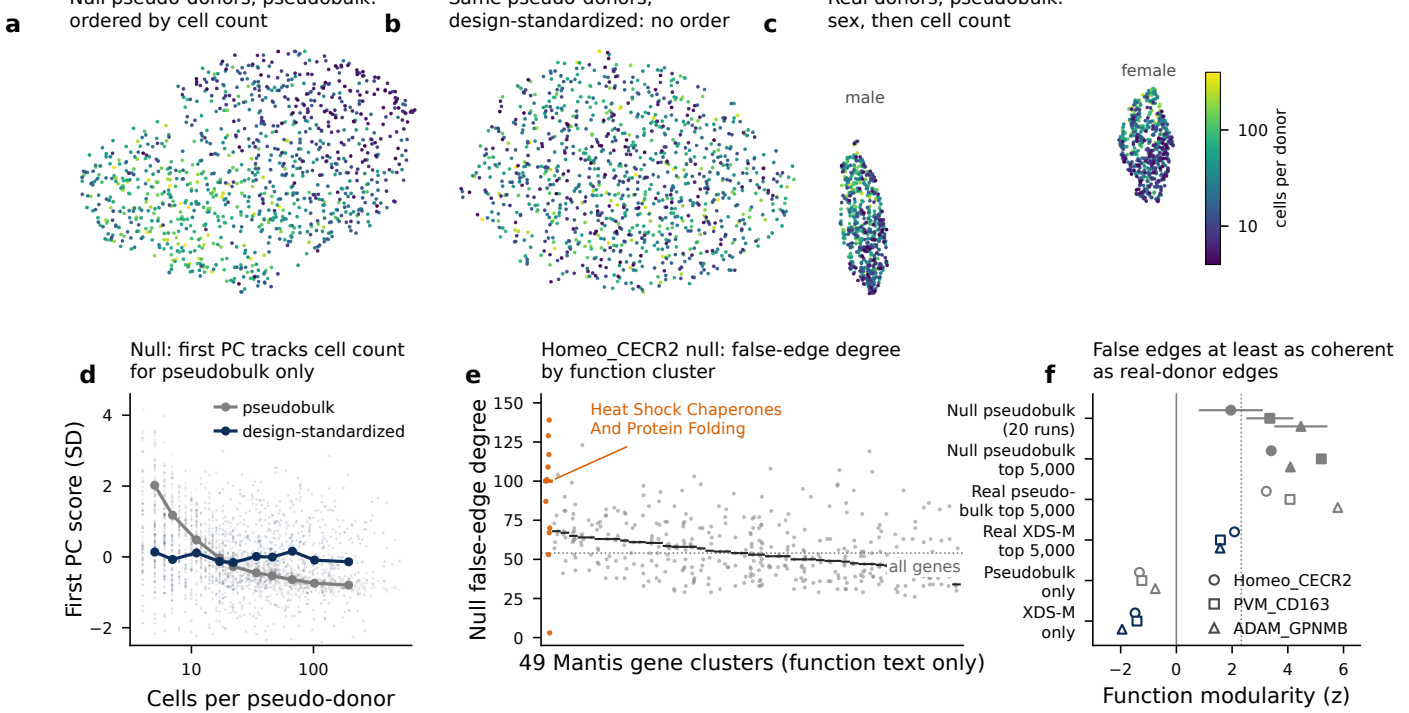

a
Null pseudo-donors, pseudobulk:
ordered by cell count
b
Same pseudo-donors,
design-standardized: no order
c
Real donors, pseudobulk:
sex, then cell count
male
female
100
10
cells per donor
d
Null: first PC tracks cell count
for pseudobulk only
pseudobulk
design-standardized
First PC score (SD)
4
2
0
−2
10
100
Cells per pseudo-donor
e
Homeo_CECR2 null: false-edge degree
by function cluster
Heat Shock Chaperones
And Protein Folding
all genes
Null false-edge degree
150
125
100
75
50
25
0
49 Mantis gene clusters (function text only)
f
False edges at least as coherent
as real-donor edges
Null pseudobulk
(20 runs)
Null pseudobulk
top 5,000
Real pseudo-
bulk top 5,000
Real XDS-M
top 5,000
Pseudobulk
only
XDS-M
only
Homeo_CECR2
PVM_CD163
ADAM_GPNMB
−2
0
2
4
6
Function modularity (z)